

Seconder of the vote of thanks to Grünwald, de Heide and Koolen and contribution to the discussion of ‘Safe Testing’

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Congratulations to Grünwald, de Heide, and Koolen (GHK) for a wonderful contribution. Based on the comments from others following GHK’s presentation, it’s clear that the paper has already had an impact, and I fully expect that impact to spread more broadly in years to come. Indeed, GHK’s paper has already inspired me, and my comments/questions below are a reflection of this inspiration.

Exact or approximate growth rate optimality? Roughly, GHK’s proposal is to define e-values (or e-processes) based on maximizing the growth rate under the alternative hypothesis. This is appealing in a number of respects, but it can fail to produce an e-process in some cases and would be non-trivial to carry out in complex, high-dimensional problems. But there are relatively simple e-process constructions, e.g., via universal inference in Wasserman et al. (2020) and Ramdas et al. (2023), that achieve a certain growth rate, but probably are not exactly optimal in GHK’s sense. In Dixit and Martin (2023) we proposed an e-process based on a nonparametric mixture model alternative fit recursively via Michael Newton’s *predictive recursion* algorithm (e.g., Martin 2021; Martin and Tokdar 2009, 2011; Newton 2002) with asymptotic growth rate comparable to GHK’s exact growth rate, expressed in terms of Kullback–Leibler projections, etc. The question I have is if there are practical or theoretical benefits to GHK’s exact growth rate optimality compared to something like this asymptotic growth rate optimality. If so, then those benefits would be worth understanding.

Safety against all or only some stopping times? An obvious criticism of GHK’s specific proposal, and safe/anytime-valid inference in general, is that the methods (tests and confidence regions) are more conservative than their classical counterparts. The equally obvious rebuttal is that these new methods offer something that the classical methods don’t—the efficiency loss is the price for validity uniformly over the often-unspecified stopping time used when collecting the data. While classical methods don’t *automatically* accommodate uncertain stopping times, they can be *manually* adjusted to do so, and this has some advantages and disadvantages as I explain below.

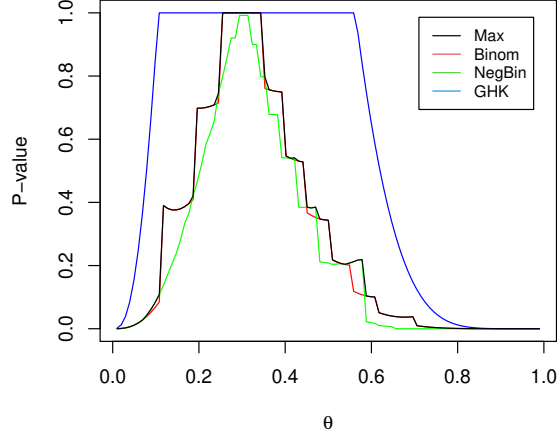


Figure 1: Plot of four different p-value functions for the Bernoulli success probability based on 10 trials and 3 successes. Red and green are for the binomial and negative binomial models, respectively, black is the pointwise maximum based on (1), and blue is the e-value based on GHK’s Example 5 converted to the p-value scale.

Suppose that the data can be represented as (N, X^N) , where N is a stopping time, and the statistical model is P_θ , where $\theta \in \Theta$ is unknown. I don’t have space here to explain, so I’ll simply refer to (1) as a “p-value function” but it’s actually the basic building block of a principled framework for frequentist reliable (imprecise) probabilistic inference (Martin 2022). For observed data (n, x^n) , define a “safe” p-value function as

$$\pi_{(n, x^n)}(\vartheta) = \sup_{N: \text{supp}(N) \ni n} P_\vartheta\{\text{rlik}(\vartheta, X^N) \leq \text{rlik}(\vartheta, x^n)\}, \quad \vartheta \in \Theta, \quad (1)$$

where $\text{rlik}(\cdot, x^n)$ is the relative likelihood function based on data x^n —which only depends on the observed x ’s, not directly on n —and the supremum is over stopping times N that could’ve taken value n , i.e., stopping rules whose support contains n . This corresponds to treating the stopping time N as a nuisance parameter (Martin 2023a,b), which I think makes sense. The advantage to this approach is that one isn’t obligated to consider *all* stopping times—the above optimization need only be carried out over the possible stopping times. So, if the data analyst can justify excluding certain stopping times from consideration, then the range of optimization can be reduced. This, in turn, makes the p-value function (1) smaller and, hence, improves efficiency without impacting safety.

A too-simple example should help to illustrate my point. Suppose that the data consists of a sequence of Bernoulli trials with unknown success probability θ . The observed data consists of 10 trials with 3 successes. If the investigator believes that there are only two possible stopping rules in consideration—fixed 10 trials or fixed 3 successes—then these correspond to a binomial and negative binomial models, respectively. Figure 1 plots the p-value function for these two models, along with the safe pointwise maximum corresponding to (1). GHK’s growth rate optimal e-value (their Example 5) is converted to a p-value (Grünwald 2023) and also displayed. Smaller p-value functions means greater efficiency, so the benefit of not considering all stopping times is clear.

The disadvantage, of course, is that the supremum over (some) stopping times in (1) could be difficult/impossible to compute in certain cases. With only finitely many

stopping times in consideration, the computations wouldn't be much more difficult than in the above example. Also, the “sup” in (1) is a very special case of a Choquet integral with respect to an imprecise prior distribution for N . I'm optimistic that this characterization, along with available computational tools for Choquet integration, can be leveraged to efficiently handle the “some stopping times” case.

To conclude, I want to congratulate GHK again for their inspiring and thought-provoking contribution. It's my pleasure to have had the opportunity to participate in the discussion of this excellent paper and, in particular, to second the vote of thanks.

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